

Twitter and the Politics of the Federal Deficit
Results Summary

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2025-10-06

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! Important

Writing sample disclaimer. This PDF contains a draft of the *Results* section from the working paper **Twitter and the Politics of the Federal Deficit**. The abstract is included below for context.

0.1 Abstract (for context)

For decades, U.S political parties circumstantially exploit fears around the the federal deficit to advance partisan goals. The shift to social-media based political communication allows us to empirically assess this long-held belief. Analyzing ~3.6 million tweets from official congressional accounts (June 2017–January 2023), I identify deficit-related posts using a two-tier keyword approach and estimate member–month logistic models linked to leadership roles and legislative windows. Deficit rhetoric is uncommon overall— **0.7%** of tweets (7 per 1,000)—but varies sharply with political context: Republicans tweet about the deficit five times as often when in the minority, while Democrats’ rates remain comparatively stable across majority/minority status. Legislative periods further amplify these patterns . The results confirm synchronized, context-responsive messaging—especially for the GOP—and show that deficit rhetoric is less a benchmark of fiscal conditions and operates as a communication-weapon serving partisan ends.

1 Macro Trends

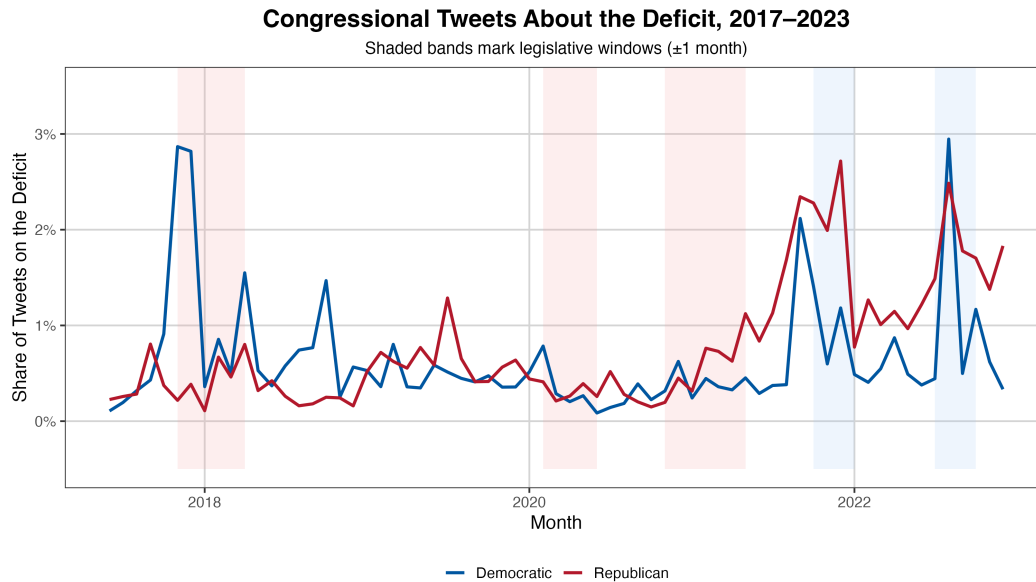


Figure 1: Share of congressional tweets about the deficit

Figure 1 shows the monthly share of congressional deficit tweets, by party, from June 2017 to January 2023. Consistent with best practices in political science, this paper will represent the Republican Party with red shading and the Democratic Party with blue. Shaded areas indicate legislative periods – defined as the month before, during, and after the passage of a fiscally significant bill. Legislative periods exclude small appropriations, procedural votes, and commemorations.

The baseline deficit tweeting share for both parties is low, generally staying under 1% averaged across members across the six years, with notable accelerations occurring around legislative periods. Democratic tweeting patterns remain relatively consistent across the six years, with three notable peaks, all during legislative periods: December 2017 (Tax Cuts and Jobs Act), October 2021 (Infrastructure Investment and Jobs Act), and August 2022 (Inflation Reduction Act, CHIPS & Science Act). Republicans, however, demonstrate consistently higher deficit tweeting rates as the minority party, as illustrated in the peaks (2022, 2023) compared to relatively low rates from 2017–2021.

2 Economic Indicators

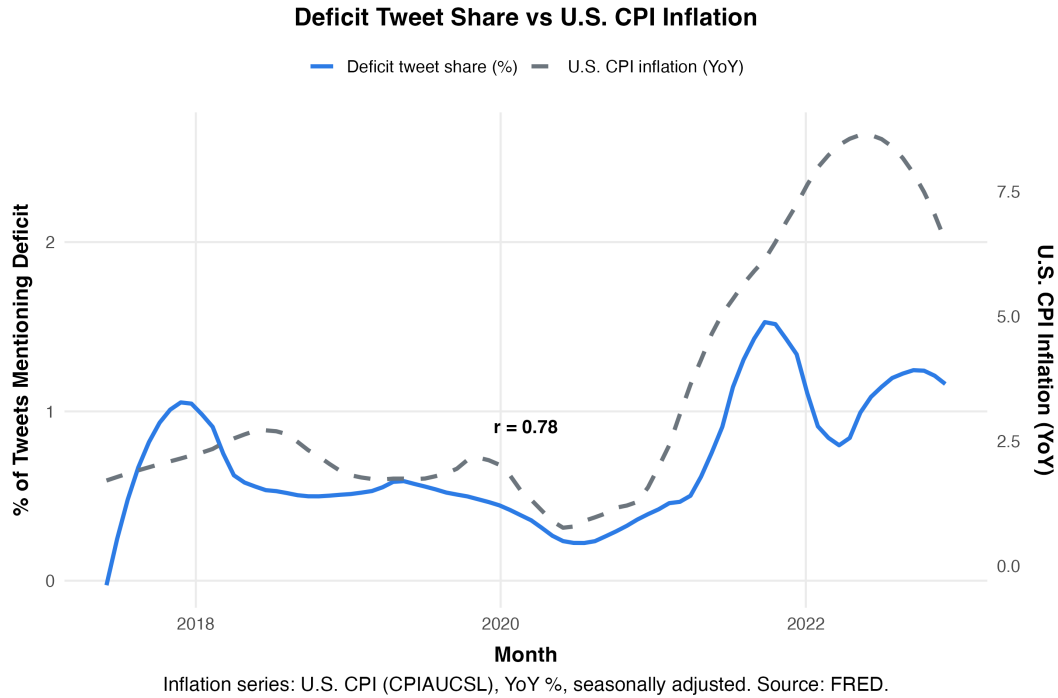


Figure 2: Deficit tweet share vs. U.S. CPI inflation.

Figure 2 plots deficit tweet share against US CPI inflation (YOY) reveals a relatively strong positive relationship, as indicated by its large r-value (**.78**). There are a few plausible explanations for this linear relationship. First, the elevated inflation rates of 2022, peaking at **9.1%** in June(40-year high), may have increased elected officials concern about deficit spending. Public discontent over higher prices may have also pressured representatives to take a more hawkish approach on federal expenditures.

Another explanation could be that along with higher inflation, the significant fiscal legislation being proposed in the summer of 2022 (eg, IRA, CHIPS) may have drawn concerns from representatives prompting more congressional tweets on the topic.

3 Member-Level Patterns

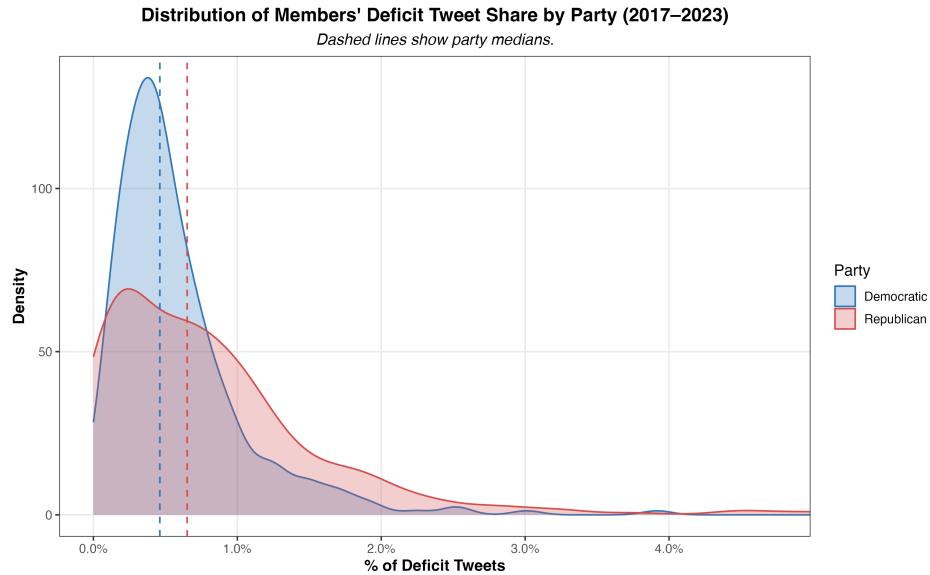


Figure 3: Share of deficit tweets by party (member-level distribution).

Figure 3 presents member-level distributions of deficit tweeting, calculated as the share of each member's tweets that mention the deficit over the entire period. As illustrated by the dashed lines (party medians), Republicans tweet about the deficit more frequently than Democrats, with average shares of **0.81%** and **0.62%**, respectively.

Democrats display a relatively compact distribution, with most representatives falling below **0.7%**, indicating less inter-party variation in deficit tweeting. Republicans, however, show far greater variability within their party. The GOP's right tail distribution illustrates that, compared to their Democratic colleagues, a far greater proportion of members tweet about the deficit at much higher rates, with outliers reaching **4–8%**.

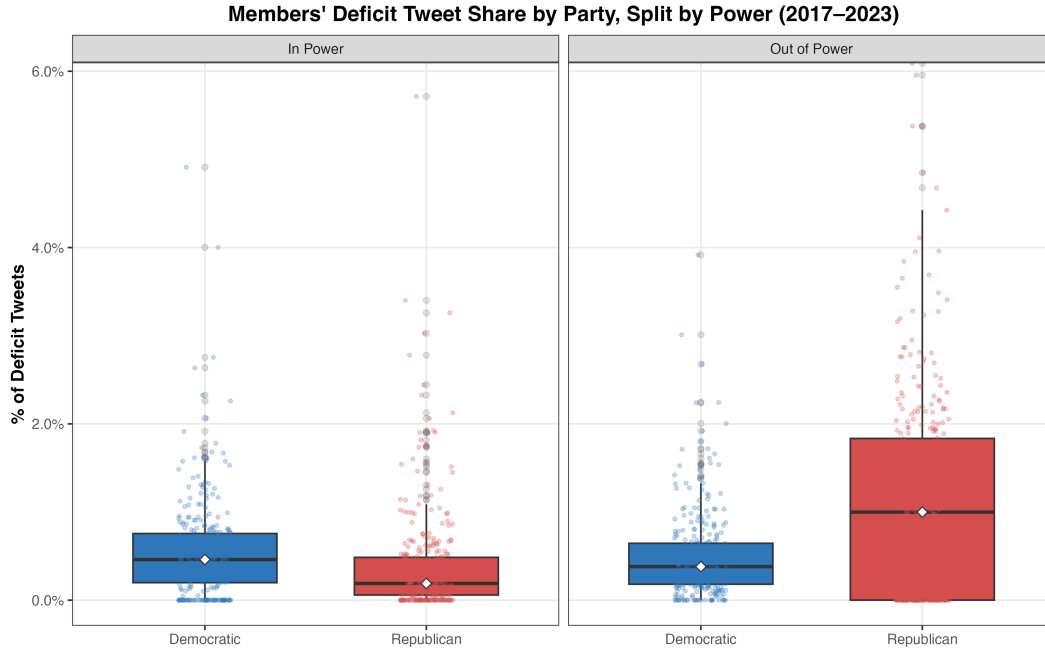


Figure 4: Distribution of deficit tweets by power (boxplot).

Figure 4 breaks down these distributions further by institutional power status (majority vs. minority). The left and right panels juxtapose the distribution when a member’s party controls the presidency (in-power) versus when it does not (out-of-power). For instance, Democrats would be considered in-power from January 2021 to January 2023 under President Joe Biden, while Republicans would be out-of-power during this same period.

Figure 4 shows a clear asymmetry in GOP tweeting patterns between these periods. When moving between power contexts, the GOP median tweet share increases five-fold **0.2% to 1.0%** and the inter-quartile range from **0.6% to 1.8%**. In other words, Republicans were far more likely to tweet about the deficit under President Biden than under President Trump. This stark distributional contrast illustrates a synchronized and cohesive shift in messaging among Republicans: emphasize the deficit more in periods of political weakness. Democrats, by contrast, remain relatively stable across contexts and actually demonstrate a modest increase in spread and averages when in power.

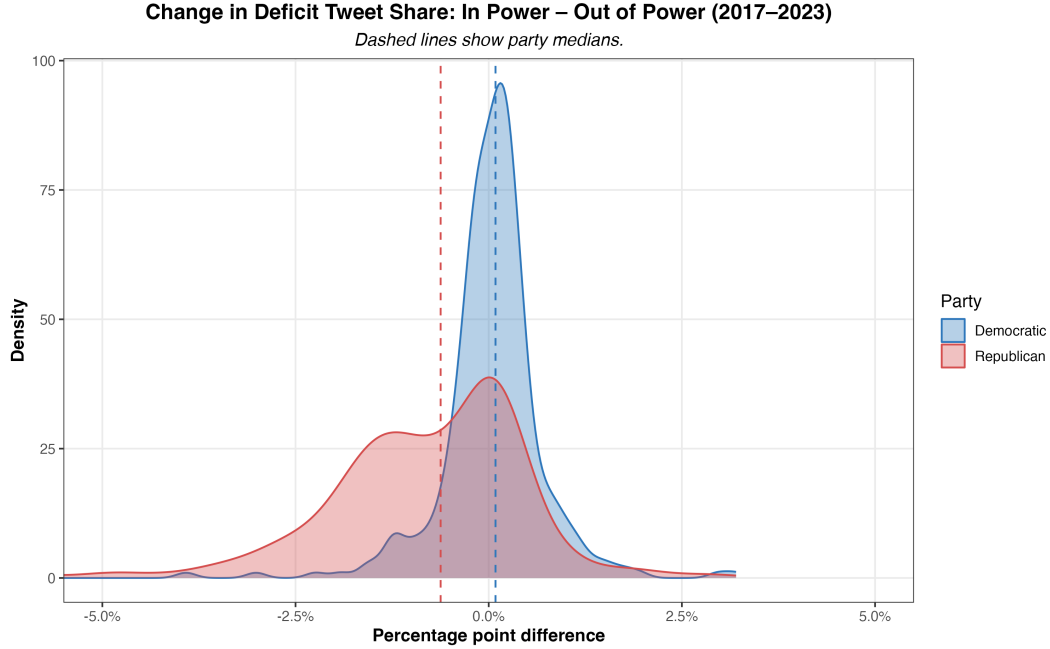


Figure 5: In-out shifts in deficit tweeting (histogram).

Figure 5 visualizes these shifts more directly by plotting the distribution of changes in deficit tweeting when moving from in-power to out-of-power contexts. The x-axis represents the difference in deficit tweet share between these two periods for each member (in minus out). Positive values indicate that a member tweeted more about the deficit when in power, while negative values indicate higher tweeting rates when out of power. The GOP’s comprehensive party shift, highlighted is evident, with the bulk of their parties distribution skewed negatively.

Taken together, Figure 4 and Figure 5 suggest that Republicans are far more sensitive to power status when deciding how frequently to tweet about the deficit, while Democrats maintain a more consistent messaging strategy regardless of institutional control. These results illustrate a broader point about the coordination of modern political communication strategies on Twitter. The pronounced and systematic increase in Republican deficit tweeting supports the notion of uniform, top-down messaging within parties.

4 Leadership Analysis

Deficit Tweeting Rates by Congressional Leaders (2017–2023)				
<i>Includes only leaders who served during this period</i>				
Leadership Role	Party	Total Tweets	Debt Tweets	% Deficit-Related
Chuck Schumer				
Senate Majority Leader	Democratic	4173	167	4.00
Senate Minority Leader	Democratic	8994	93	1.03
Kevin McCarthy				
House Minority Leader	Republican	762	0	0.00
Mitch McConnell				
Senate Majority Leader	Republican	5622	3	0.05
Senate Minority Leader	Republican	2483	47	1.89
Nancy Pelosi				
House Majority Leader	Democratic	5617	28	0.50
House Minority Leader	Democratic	2705	30	1.11
Paul Ryan				
House Majority Leader	Republican	3570	6	0.17

Figure 6: Deficit tweeting by congressional leaders (House and Senate; majority vs. minority roles).

Figure 6 summarizes deficit tweeting among House and Senate leaders in both majority and minority roles. Tweeting patterns vary by party and institutional position, with the sharpest contrasts appearing in the Senate. Senator Chuck Schumer tweeted about deficits far more often as Majority Leader **4.0%** than as Minority Leader **1.0%**, while Senator Mitch McConnell showed the reverse, posting more in the minority **1.9%** than the majority **0.05%**. Notably, in contrast to Senate counterpart, Speaker Nancy Pelosi displayed higher rates in the minority **1.1%** than as Speaker **0.5%**. Representative McCarthy's abnormally small number of registered tweets (762), suggest a potential issue in reconciling changes in his twitter handle. Considering congressional leaders often maintain multiple twitter accounts, with new handles introduced and changed depending on their relative power status (majority, minority, rank-and-file), some of Representative McCarthy's twitter accounts may have not been picked up by the data.

Representative Paul Ryan - a self-proclaimed "deficit hawk" and speaker of the House from 2017-2019 - is an central political figure in discourse surrounding the national deficit. Speaker Ryan, whose deficit tweet share is just **0.17%**, has been long criticized for his messaging around the national debt - which often conflict with his historical voting patterns towards fiscally significant bills.

A 2018 Politico op-ed *Paul Ryan's Legacy of Red Ink* writes "Ryan loves to *talk* about reducing the national debt, but what he loves to *do* is reduce taxes, which increases the national debt". My research highlights an important nuance to this point. While Speaker Ryan may emphasize the deficit in out-of-power contexts, he does not tweet about the deficit in periods of legislative control. Ryan's deficit apprehensions may be due to his legislative agenda during this period, passing the Tax Cuts and Jobs Act (2017), which, by CBO estimates, will balloon the national debt by \$1.5 Trillion over the following decade. Further analysis of Speaker Ryan's tweeting habits, especially in instances of minority power, would help clarify his communication patterns, and the degree to which he uses the federal deficit as a tool for partisan ends. Nonetheless, this research

confirms an important point about the messaging strategy of a key American legislator: Paul Ryan, a self-identified “fiscal conservative”, rarely mentioned the deficit when holding complete legislative control from 2017 to 2019.

Aside from highlighting rhetoric patterns in key legislators, @fig-leadership also underscores a broader point about communication uniformity among party leaders and rank-and-file representatives. Previous research establishes the growing polarization in congressional ideology and party-line voting over recent decades (CQ Roll Call 2024). My research provides insight into the specific polarities of a single rhetorical subject - the federal deficit.

For Democrats, the divergent tweeting patterns of Speaker Pelosi and Senator Schumer suggest a comparatively weaker inter-party alignment in messaging. These patterns are consistent with the broader Democratic caucus, which show no discernible difference in deficit tweeting between power contexts (as shown in Figure 4 and Figure 5).

The GOP, however, presents a more uniform party dynamic. Senator McConnell’s significant increase in deficit tweeting in out of power contexts mirrors the broader Republican trend illustrated in Figure 5. The two messaging strategies employed by party leaders reflect a broader division in party dynamics between Democrats and the GOP. Democratic leaders’ deficit messaging is not tied to or correlated with their power status within the party. Republicans, by contrast, demonstrate higher levels of systematic and uniform shifts in messaging between power contexts.

5 Legislative Context

Deficit Share Within Each Window by Party (Tweet Level)				
Conditional share of deficit tweets given the tweet is inside the window.				
		Tweets (N)	Deficit (N)	Deficit Rate (%)
All Tweets (Baseline)	Democratic	2,138,163	13,199	0.6%
	Republican	1,467,248	11,903	0.8%
Any Legislative Window	Democratic	687,875	5,569	0.8%
	Republican	486,104	4,558	0.9%
IRA / CHIPS Window	Democratic	91,767	1,194	1.3%
	Republican	74,031	1,421	1.9%
CARES Window	Democratic	131,549	521	0.4%
	Republican	82,587	231	0.3%
PPP–HCE Window	Democratic	135,059	342	0.3%
	Republican	88,440	250	0.3%
COVID Window (CARES or PPP–HCE)	Democratic	171,383	627	0.4%
	Republican	109,109	335	0.3%
TCJA Window	Democratic	85,463	1,680	2.0%
	Republican	62,372	139	0.2%

*Percentages are conditional on being inside each window; counts are tweet totals **within** the window.*

Figure 7: Deficit share within legislative windows, by party.

Legislative periods – defined by one month before and after bill passage – raise the conditional rate modestly to **(0.9%)**. Industrial policy bills – the Inflation Reduction Act (IRA) and CHIPS Act, passed under President Biden – generated the highest levels of deficit tweeting **(1.6%)** among passed legislation. The Tax Cuts and Jobs Act (2017), passed under President Trump, had a deficit tweet share of **(1.2%)** – higher than baseline, but lower than the IRA/CHIPS.

@fig-deficit-share-by-party decomposes these differences by party. At baseline, Republicans tweet about the deficit more often **(0.8%)** than Democrats **(0.6%)** – a gap that persists across most legislative windows. During the IRA and CHIPS periods, Republicans’ tweets accelerated, with deficit mentions at nearly **(2.0%)**, compared to **(1.3%)** among Democrats. However, the TCJA – a partisan GOP tax cut– is a noticeable and stark exception. During these three months, the Democrats’ deficit share spikes to **(2.0%)**, while the Republicans’ share falls to just **(0.2%)**, reflecting a partisan realignment around a highly divisive bill. In contrast, during the COVID relief period in 2020 – a non-partisan voting context – both parties converged at very low levels **(0.3–0.4%)**.

6 Regression Visualization

The following regression models offer a structured approach to examining how institutional, fiscal, and partisan contexts affect the likelihood of congressional deficit tweeting.

Since the specifications rely heavily on interaction terms, it is useful to provide a framework for their interpretation first. The coefficients are expressed as odds ratios: values above one indicate that Republicans are more likely to tweet about deficits relative to Democrats. In contrast, values below one indicate a narrowing of the GOP-Dem gap. Each model includes majority interaction coefficients of the general form $\text{GOP} \times \text{Conditioning Variable}$. These coefficients indicate whether the baseline Republican–Democratic difference changes under specific conditions (legislative windows, presidential control, chamber control, or trifectas).

Legislative Windows and Major Bills		
	Odds Ratio	95% CI
GOP × Legislative window	0.88	[0.57, 1.36]
GOP × COVID window	0.65	[0.33, 1.28]
GOP × IRA/CHIPS window	1.24	[0.46, 3.32]
GOP × Bill partisanship (z)	0.82	[0.52, 1.29]
GOP × Fiscal magnitude (z)	1.02	[0.97, 1.07]

Republican–Democrat differences in the odds of tweeting about the deficit across legislative windows and major-bill periods, with member and month fixed effects.

- Odds ratios with 95% CIs; stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
- Member and calendar-month fixed effects; SEs two-way clustered by member and month.
- Bill partisanship (z): 1 SD increase in mean bill partisanship score.
- Fiscal magnitude (z): 1 SD increase in total 10-year deficit impact.
- COVID window and IRA/CHIPS window are month-level indicators.

Figure 8: Regression 1: Legislative windows.

The baseline model, Figure 8, examines partisan differences in legislative periods without conditioning on institutional control. The non-significant results suggest that, when averaged across both Democratic and Republican presidencies, legislative windows themselves do not systematically impact the partisan gap in deficit tweeting.

Presidential Control and Legislative Windows		
	Odds Ratio	95% CI
GOP × Legislative window	0.38***	[0.26, 0.56]
GOP × Minority presidency	1.87*	[1.07, 3.27]
GOP × Legislative window × Minority presidency	4.24***	[2.27, 7.92]
GOP × COVID window	1.03	[0.61, 1.74]
GOP × IRA/CHIPS window	0.78	[0.33, 1.85]
GOP × Bill partisanship (z)	0.51***	[0.41, 0.63]
GOP × Fiscal magnitude (z)	0.99	[0.98, 1.01]

Coefficients are GOP–Dem gaps: outside-leg gap under minority presidency; the legislative-window gap under same-party presidency; and how that gap changes under minority presidency.

- Odds ratios with 95% CIs; stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
- Member and calendar-month fixed effects; SEs two-way clustered by member and month.
- Bill partisanship (z) and fiscal magnitude (z) are standardized.

Figure 9: Regression 2: Majority context (presidency).

Figure 9, which controls for presidential party, substantially changes the results from the preceding specification.

Republicans’ odds of tweeting about deficits are nearly double (**OR = 1.87**) when not holding the presidency compared to Democrats under the same condition. In contrast, **GOP × Legislative window (OR = 0.38)** highlights that under a Republican presidency, the partisan gap narrows during legislative months; the odds of the GOP tweeting are **62%** lower than those of Democrats. This pattern aligns with Figure 1, which shows that Republican tweeting contracts while Democratic tweeting spikes as Congress passes the Tax Cuts and Jobs Act in late 2017. However, the three-way interaction coefficient **GOP × Legislative window × Minority presidency** reveals a reversal of this trend (**OR = 4.24**). Under a Democratic presidency, Republicans’ odds of tweeting are more than four times higher than those of Democrats in the same condition.

Government Control and Legislative Windows		
	Odds Ratio	95% CI
GOP × Legislative window (GOP trifecta)	0.31***	[0.21, 0.46]
GOP × Legislative window × Dem trifecta	7.00***	[4.13, 11.85]
GOP × Legislative window × GOP Senate+Presidency	0.85	[0.47, 1.54]
GOP × COVID window	1.10	[0.62, 1.95]
GOP × IRA/CHIPS window	0.79	[0.33, 1.86]
GOP × Bill partisanship (z)	0.50***	[0.39, 0.64]
GOP × Fiscal magnitude (z)	0.99	[0.98, 1.01]

Baseline: Democrats vs Republicans during legislative windows under a GOP trifecta. Rows report how that GOP–Dem gap changes under alternative government control states. Pure Democrat shifts and a baseline GOP main effect are omitted by design.

- Odds ratios with 95% CIs; stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.
- Member and month fixed effects; SEs two-way clustered by member and month.
- Bill partisanship (z) and fiscal magnitude (z) are standardized.

Figure 10: Regression 3: Majority context — chamber and presidency combinations.

The third model further examines the role of institutional power by including combinations of chamber control. The regression baseline is Democrats, outside legislative windows, when their party controls nothing (i.e., a minority in the House and Senate; does not hold the Presidency).

GOP × Legislative window (OR = 0.31) indicates that during legislative months, when holding a trifecta, Republicans’ odds of tweeting are **69%** lower than Democrats. The Tax Cuts and Jobs Act (2017) – a legislative window under a GOP trifecta – is an illustration of this circumstance. Once again, once we condition on a Democratic trifecta, this effect reverses. Relative to the baseline condition (a GOP trifecta), Republicans’ odds of deficit tweeting are seven times higher than those of Democrats in the same context. The juxtaposition of these two coefficients (**OR = 0.31**) versus (**OR = 7.00**) illustrates that legislative timing by itself isn’t determinative of tweeting gaps – but who governs is. Under Democratic control (i.e.,

2021-2023), Republicans accelerate deficit rhetoric during legislative months, while under the GOP trifecta (i.e., 2017-2019), they significantly reduce deficit tweeting frequencies.

6.1 Bill Partisanship and Fiscal Magnitude

Across all three specifications, the coefficients for Fiscal magnitude (z) remain consistently insignificant (~ 1.0). These coefficients, based on a normalized measure of a given bill's 10-year CBO estimate, suggest that the fiscal consequences of policy do not systematically impact partisan tweeting gaps.

Bill partisanship (z), however, shows significant contractions in the GOP-Dem tweeting gap once institutional control is included (**OR** $\sim$ **0.50**). In other words, for each one-SD increase in a bill's partisanship score, the GOP-Dem gap contracts by roughly half. Notably, because these variables are continuous (ie, z-scores), the coefficients represent slopes rather than a baseline-specific effect. For example, in the context of bill partisanship, if Republicans are twice as likely as Democrats to tweet at $\mathbf{z} = \mathbf{0}$ (**OR** = **2.0**), that gap narrows to parity at $\mathbf{z} = \mathbf{1}$ (**OR** = **1.0**).